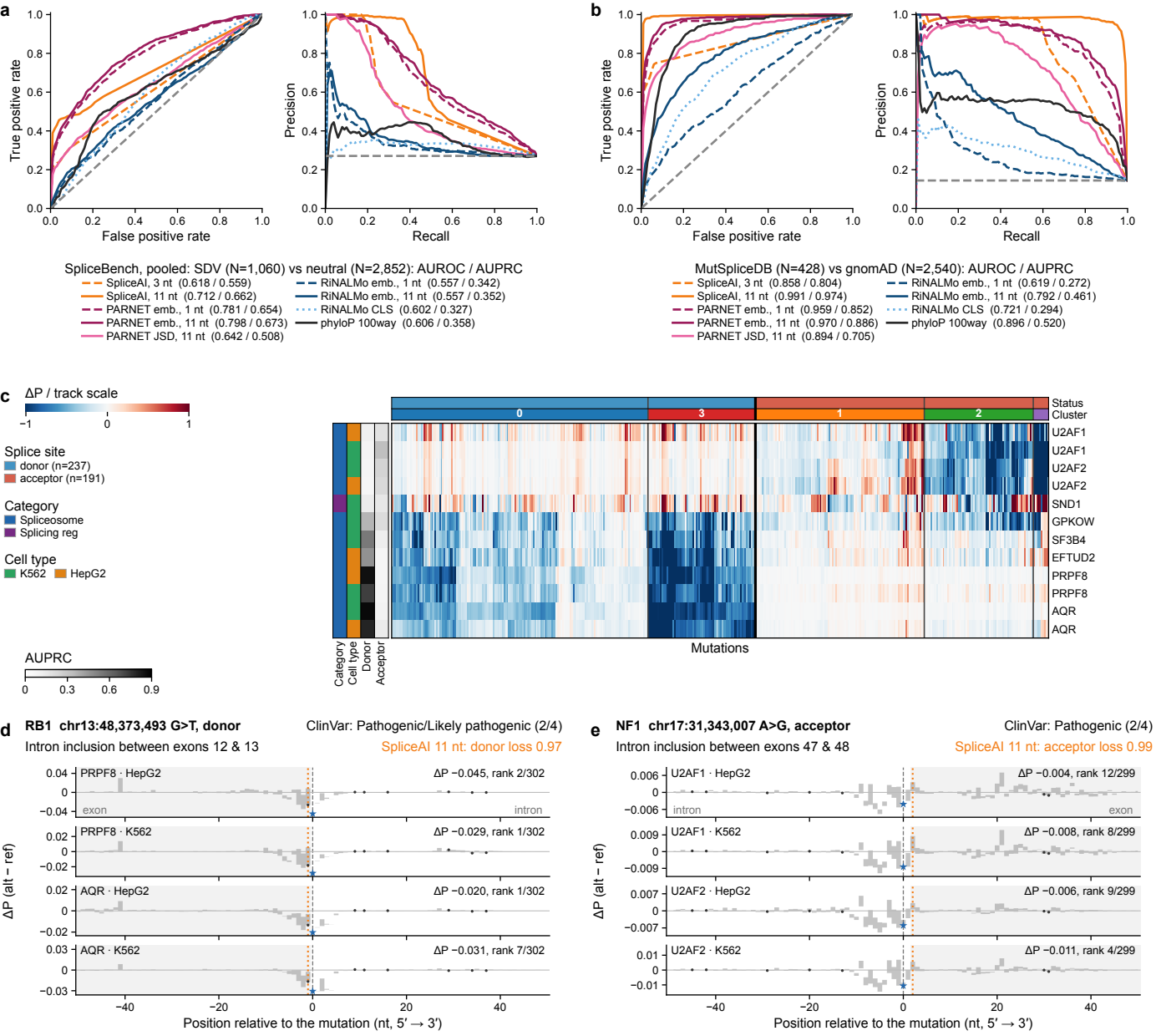

a
True positive rate
False positive rate
Precision
Recall
SpliceBench, pooled: SDV (N=1,060) vs neutral (N=2,852): AUROC / AUPRC
SpliceAI, 3 nt (0.618 / 0.559)
SpliceAI, 11 nt (0.712 / 0.662)
PARNET emb., 1 nt (0.781 / 0.654)
PARNET emb., 11 nt (0.798 / 0.673)
PARNET JSD, 11 nt (0.642 / 0.508)
RiNALMo emb., 1 nt (0.557 / 0.342)
RiNALMo emb., 11 nt (0.557 / 0.352)
RiNALMo CLS (0.602 / 0.327)
phyloP 100way (0.606 / 0.358)
b
MutSpliceDB (N=428) vs gnomAD (N=2,540): AUROC / AUPRC
SpliceAI, 3 nt (0.858 / 0.804)
SpliceAI, 11 nt (0.991 / 0.974)
PARNET emb., 1 nt (0.959 / 0.852)
PARNET emb., 11 nt (0.970 / 0.886)
PARNET JSD, 11 nt (0.894 / 0.705)
RiNALMo emb., 1 nt (0.619 / 0.272)
RiNALMo emb., 11 nt (0.792 / 0.461)
RiNALMo CLS (0.721 / 0.294)
phyloP 100way (0.896 / 0.520)
c
ΔP / track scale
Splice site
donor (n=237)
acceptor (n=191)
Category
Spliceosome
Splicing reg
Cell type
K562
HepG2
AUPRC
Status
Cluster
U2AF1
U2AF1
U2AF2
U2AF2
SND1
GPKOW
SF3B4
EFTUD2
PRPF8
PRPF8
AQR
AQR
Mutations
Category
Cell type
Donor
Acceptor
d
RB1 chr13:48,373,493 G>T, donor
ClinVar: Pathogenic/Likely pathogenic (2/4)
Intron inclusion between exons 12 & 13
SpliceAI 11 nt: donor loss 0.97
PRPF8 · HepG2
ΔP −0.045, rank 2/302
exon
intron
PRPF8 · K562
ΔP −0.029, rank 1/302
AQR · HepG2
ΔP −0.020, rank 1/302
AQR · K562
ΔP −0.031, rank 7/302
ΔP (alt − ref)
Position relative to the mutation (nt, 5′ → 3′)
e
NF1 chr17:31,343,007 A>G, acceptor
ClinVar: Pathogenic (2/4)
Intron inclusion between exons 47 & 48
SpliceAI 11 nt: acceptor loss 0.99
U2AF1 · HepG2
ΔP −0.004, rank 12/299
intron
exon
U2AF1 · K562
ΔP −0.008, rank 8/299
U2AF2 · HepG2
ΔP −0.006, rank 9/299
U2AF2 · K562
ΔP −0.011, rank 4/299
ΔP (alt − ref)
Position relative to the mutation (nt, 5′ → 3′)